

SnapRedact AI

On-Device PII Redaction & Document Privacy Tool

A V U Pushyami Reddy

pushyamireddy4533@gmail.com

Snapdragon AI Lab Build & Present Challenge 2026

Powered by Snapdragon NPU | Qualcomm AI Hub

The Problem

The PII Challenge

- Organizations handle thousands of sensitive documents daily
 - Employee records, customer data, medical forms, financial docs
- Before sharing, PII must be identified and removed

Current Solutions Have Critical Flaws

- Cloud-based tools -> Data leaves your device -> Privacy risk
- Manual redaction -> Slow, expensive, error-prone
- Most tools don't support Indian PII (Aadhaar, PAN)

India's DPDPA 2023 mandates strict personal data handling.

Our Solution: SnapRedact AI

AI-powered PII detection & redaction - 100% on your Snapdragon laptop

- Zero data leakage - nothing leaves your device
- Dual AI engine - BERT NER + Smart Regex patterns
- India-first - detects Aadhaar, PAN, Indian phone numbers
- Multi-format - handles TXT, PDF, DOCX, PNG, JPG files
- Fast - Snapdragon NPU-accelerated inference
- OCR - reads scanned documents and images
- Visual dashboard with real-time detection statistics
- Export redacted docs in multiple formats

How It Works

Pipeline: Document -> Text -> PII Detection -> Redaction -> Export

Two Detection Methods Combined:

- BERT NER (AI Model) - dslim/bert-base-NER
 - Understands language context
 - Detects: Person Names, Organizations, Locations
- Regex Pattern Engine
 - Matches structured PII formats
 - Detects: Aadhaar, PAN, Email, Phone, Credit Card, DOB, IP

Results merged and deduplicated for comprehensive coverage.

Snapdragon NPU Optimization

Why On-Device AI on Snapdragon?

- ONNX Runtime -> Automatically leverages Snapdragon NPU
- On-device BERT inference -> Zero cloud API calls
- Low latency -> Near real-time detection for large documents
- Works completely offline -> No internet dependency

Privacy + Performance = Snapdragon's Unique Value

- Process sensitive documents on a flight
- Work in remote areas with no connectivity
- Comply with data localization requirements

Indian PII Detection

Built for India - 11+ PII Types Detected

- Aadhaar Number -> XXXX XXXX XXXX (12 digits)
- PAN Number -> ABCDE1234F (5 letters + 4 digits + 1 letter)
- Indian Phone -> +91-XXXXXXXXXX, 0XX-XXXXXXXX
- Indian Passport -> X1234567 (letter + 7 digits)

Plus Universal PII:

- Email addresses, Credit card numbers (16-digit)
- Dates of birth (DD/MM/YYYY), IP Addresses (IPv4)
- Organization names, Person names, Locations (via NER)

Technical Stack

Core AI

- AI Model: BERT NER (dslim/bert-base-NER) - 110M params
- Runtime: ONNX Runtime with Snapdragon NPU acceleration
- NLP: Hugging Face Transformers library

Document Processing

- OCR: Tesseract OCR + PIL image preprocessing
- PDF: PyMuPDF (text extraction + image rendering)
- DOCX: python-docx (XML parsing)

Application

- UI: Streamlit (Python web framework)
- Language: Python 3.11+
- Deployment: pip install + streamlit run (2 commands)

Application Demo

Beautiful, Interactive Dashboard

- Upload files via drag-and-drop or paste text directly
- Real-time statistics: Total PII found, AI vs Pattern counts
- Detailed entity table with type, confidence, and source
- Side-by-side: Original vs Redacted comparison
- One-click export: TXT, DOCX, or JSON report

Premium Dark Theme UI

- Gradient accents and animated hover effects
- Color-coded PII type tags for instant visual scanning
- Responsive layout works on any screen size

(See live demo or screenshots in GitHub repository)

Impact & Future Roadmap

Impact

- Compliance: Helps organizations comply with DPDPA 2023, GDPR
- Privacy: Demonstrates on-device AI for sensitive data tasks
- Accessibility: Zero cost, no cloud account, works offline

Future Roadmap

- Image PII redaction - visual masking on scanned documents
- Multi-language support - Hindi, Telugu, Tamil NER models
- Batch processing for enterprise document workflows
- ONNX model export via Qualcomm AI Hub for direct NPU execution
- FastAPI endpoint for integration with enterprise tools
- Snapdragon mobile deployment for on-the-go redaction

SnapRedact AI

Your documents, your device, your privacy.

A V U Pushyami Reddy

pushyamireddy4533@gmail.com

Snapdragon AI Lab Build & Present Challenge 2026